Assignment 6: Exploring AI Artistry: Comparing Dall-E, Midjourney, Playground.ai, and DreamStudio

Ideal solution:

Visit the websites:

Playground ai: https://playground.com/create

Midjourney ai : midjourney

- Generate 2-3 images on each platform with a prompt of your choice. Eg. lion in a cage.
- Create a doc file and note down what difference you found in midjourney vs playground.ai
- Explore how it works: go to https://openai.com/index/dall-e/ and try to understand the working.

This is a research based assignment hence there is no coding required in this assignment.

Assignment 7: Interactive Image Magic: Enhancing Flask with Dall-E

Solution:

main.py

```
from boltiotai import openai
import os
from flask import Flask, render_template_string, request

openai.api_key = os.environ['OPENAI_API_KEY']
```

```
def generate_tutorial(components):
    response = openai.Images.create(
        prompt=components,
        model="dall-e-3",
        size="1024x1024",
        response_format="url"
    )
    image_url = response['data'][0]['url']
    return image_url

app = Flask(__name__)

@app.route('/', methods=['GET'])
def hello():
    return render_template_string('''
        <!DOCTYPE html>
        <html>
        <head>
            <title>Infinite Image Generator</title>
            <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/css/bootstrap.min.css"
rel="stylesheet">
            <style>
                /* Your CSS styles go here */
            </style>
        </head>
        <body>
            <div class="container">
                <h1 class="my-4">Custom Image Generator</h1>
                <div class="interaction-container">
                    <form class="tutorial-form mb-3">
                        <div class="mb-3">
                            <label for="components" class="form-label">Textual Description of the
Image:</label>
```

```
                            <input type="text" class="form-control" name="components" placeholder="Enter the
Description (Ex: A Lion in a Cage)" required>
                        </div>
                        <button type="submit" class="btn btn-primary">Share with the Image</button>
                    </form>
                </div>
            </div>
            <script>
                async function generateImage(form) {
                    const components = form.querySelector('input[name="components"]').value;
                    const interactionContainer = form.closest('.interaction-container');

                    const outputDiv = document.createElement('div');
                    outputDiv.className = 'card mb-3';
                    outputDiv.innerHTML = `
                        <div class="card-header d-flex justify-content-between align-items-center">
                            Output:
                            <button class="btn btn-secondary btn-sm"
onclick="copyToClipboard(this)">Copy</button>
                        </div>
                        <div class="card-body">
                            <p style="white-space: pre-wrap;">Generating an image for you...</p>
                            <img style="width: auto; height: 300px;">
                        </div>
                    `;

                    const newForm = document.createElement('form');
                    newForm.className = 'tutorial-form mb-3';
                    newForm.innerHTML = `
                        <div class="mb-3">
                            <label for="components" class="form-label">Textual Description of the
Image:</label>
                            <input type="text" class="form-control" name="components" placeholder="Enter the
```

```
Description (Ex: A Lion in a Cage)" required>
                        </div>
                        <button type="submit" class="btn btn-primary">Share with the Image</button>
                    `;

                    newForm.addEventListener('submit', function(event) {
                        event.preventDefault();
                        generateImage(newForm);
                    });

                    interactionContainer.appendChild(outputDiv);
                    interactionContainer.appendChild(newForm);

                    const formData = new FormData(form);
                    const response = await fetch('/generate', {
                        method: 'POST',
                        body: formData
                    });
                    const imageUrl = await response.text();

                    outputDiv.querySelector('p').textContent = '';
                    outputDiv.querySelector('img').src = imageUrl;
                }

                function copyToClipboard(button) {
                    const imgElement = button.closest('.card').querySelector('img');
                    const imageUrl = imgElement.src;
                    const textarea = document.createElement('textarea');
                    textarea.value = imageUrl;
                    document.body.appendChild(textarea);
                    textarea.select();
                    document.execCommand('copy');
                    document.body.removeChild(textarea);
```

```
                    alert('Copied to clipboard');
                }

                document.querySelector('.tutorial-form').addEventListener('submit', function(event) {
                    event.preventDefault();
                    generateImage(this);
                });
            </script>
        </body>
        </html>
    ''')

@app.route('/generate', methods=['POST'])
def generate():
    components = request.form['components']
    return generate_tutorial(components)

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=8080)
```